Photodetection
Statistical fluctuation



October/2026
NOMACHI, Masaharu

Light detection





How do we measure weak light?
How much energy do we need to measure?

Quantized Energy flow

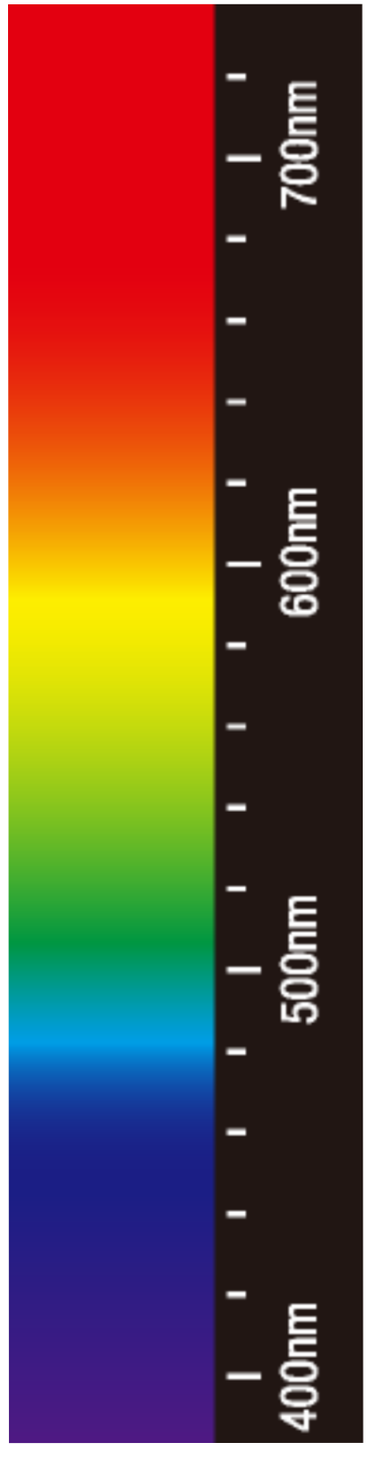
Newton and Huygens

Is light a wave or a particle?



Light behaves as a wave.

What is the medium of a light wave?
How does light travel in a vacuum?



Quantized Energy flow



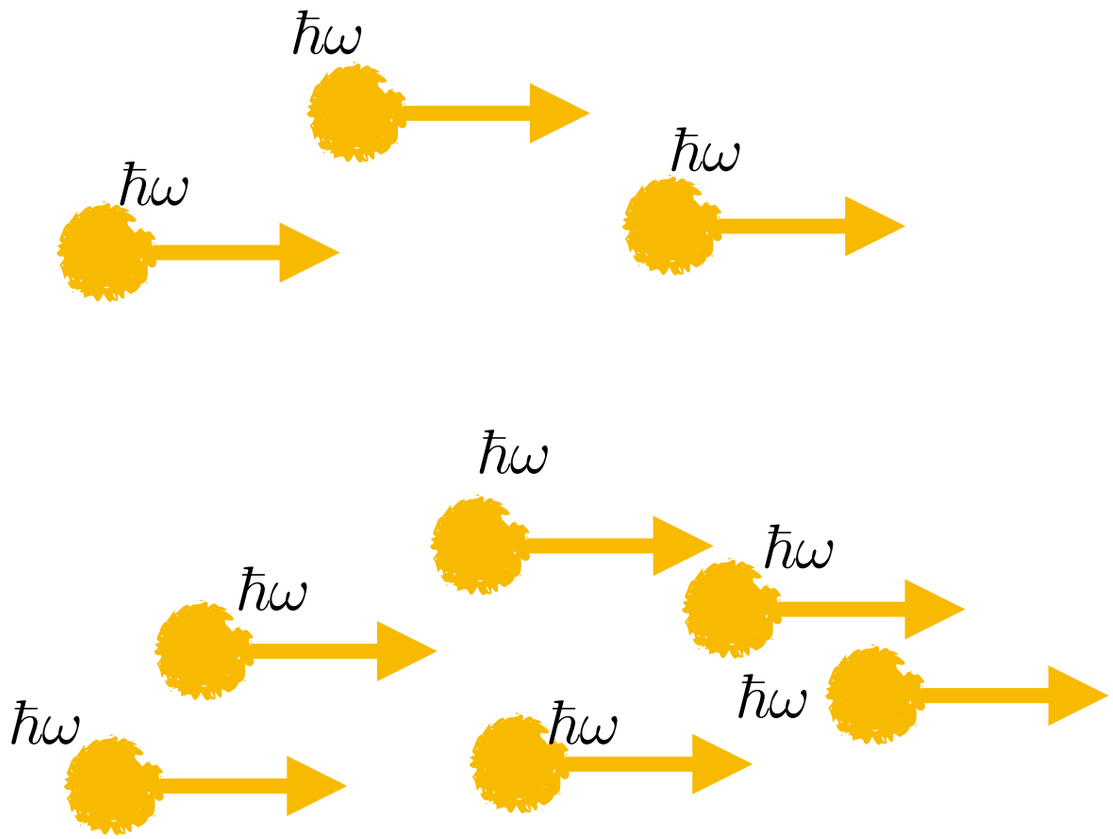
Photo from the Nobel Foundation archive.

[Max Planck](#)

The Nobel Prize in Physics 1918

Prize motivation: "in recognition of the services he rendered to the advancement of Physics by his **discovery of energy quanta**"

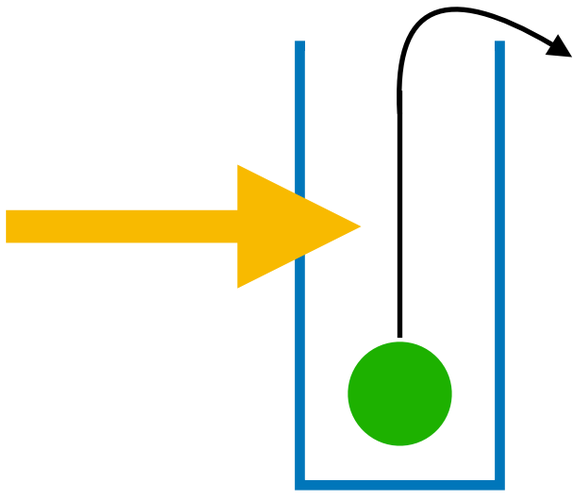




Energy flow is quantized.

Intensity is determined by the number of photons.

Color is determined by the energy of each photon.



How can we detect light

We need ENERGY to excite a microscopic system

no excitation -> no change!

How can we detect weak light

Do we need to accumulate enough energy to cause excitation?

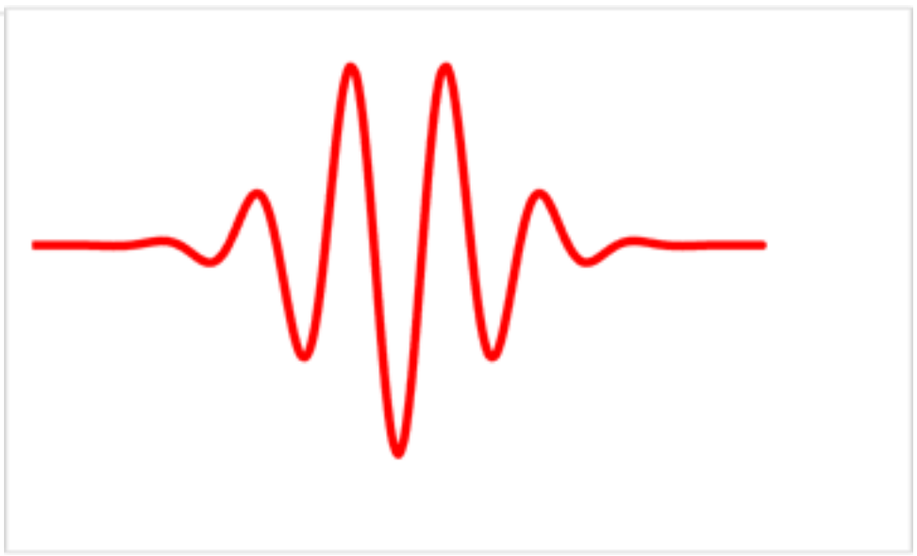


No.

Energy is quantized into photons.

Intensity = the number of photon

Count the photons.



photoelectric effect

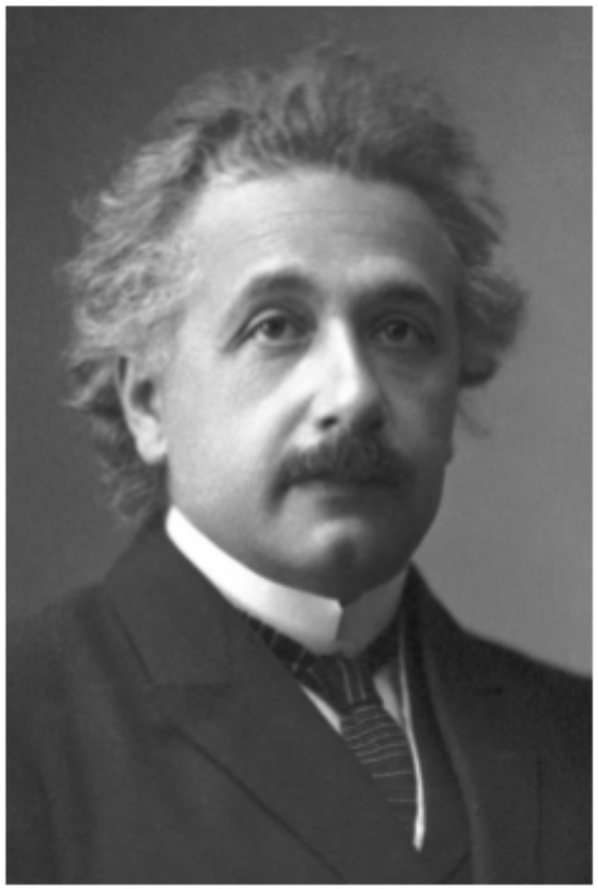
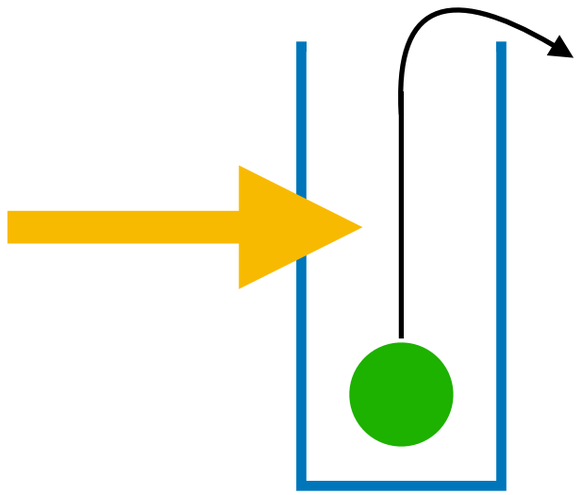


Photo from the Nobel Foundation archive.

Albert Einstein

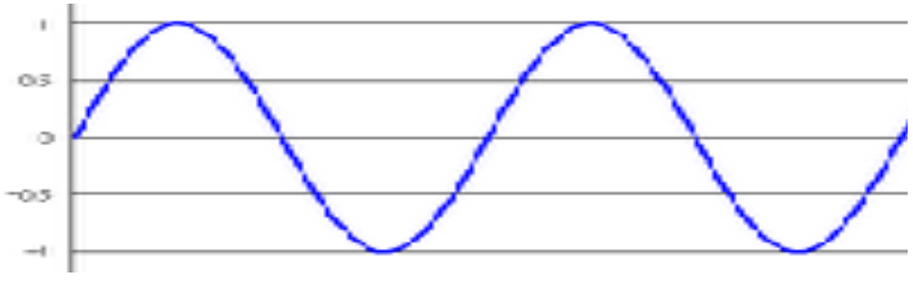
The Nobel Prize in Physics 1921

Prize motivation: "for his services to Theoretical Physics, and especially for his discovery of the law of the photoelectric effect"

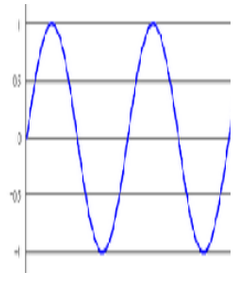


The Photon energy must exceed the Binding energy

$\hbar c$ = 197 eV nm



$\hbar c$ = 197 MeV fm



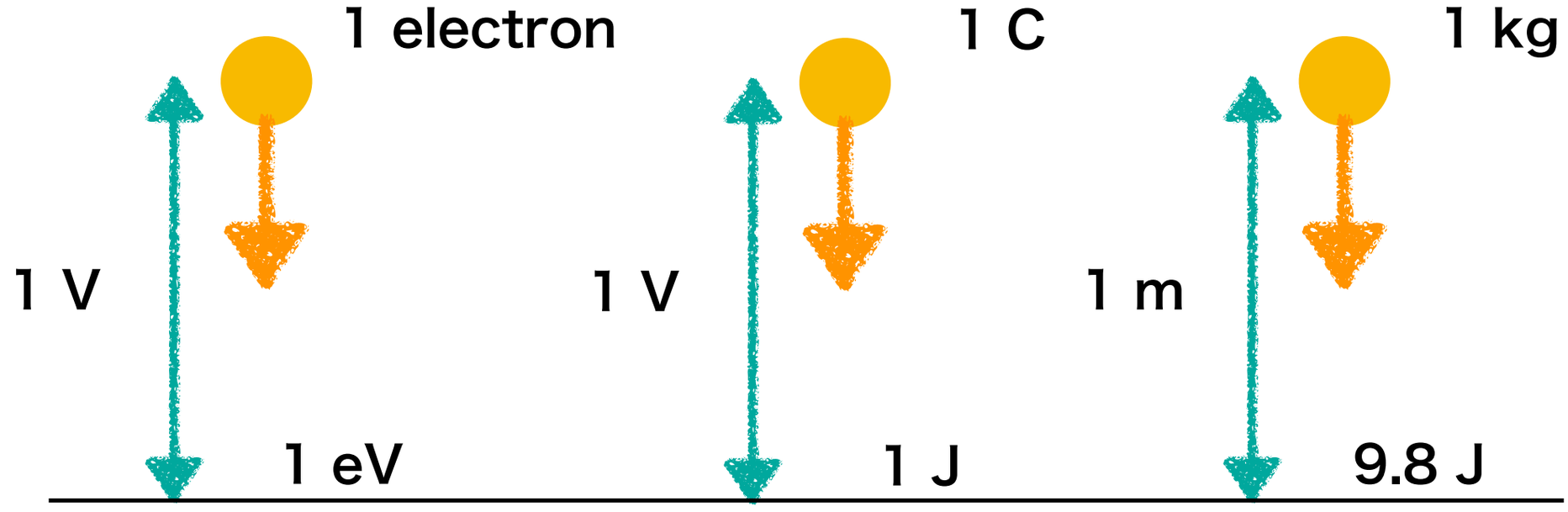
$$\text{Energy} = \frac{2\pi\hbar c}{\text{Wave length}}$$

2 eV = 600 nm (Visible light)

1 keV = 1.2 nm (～size of atom)

20MeV = 60 fm (～size of atomic nuclei)

Energy scale



$$1\ eV = 1.6 \times 10^{-19}\ J$$

chemical energy

Battery voltage is related to the chemical interactions.



Alkaline manganese battery (1.5V)

$$MnO_2 + H_2O + Zn \rightarrow Mn(OH)_2 + ZnO$$

Ni-Cd battery (1.2V)

$$2NiOOH + Cd + 2H_2O \rightleftharpoons 2Ni(OH)_2 + Cd(OH)_2$$

It is of the same order as ionization energies.

Thermal Energy

The energy of gas molecules is proportional to temperature.

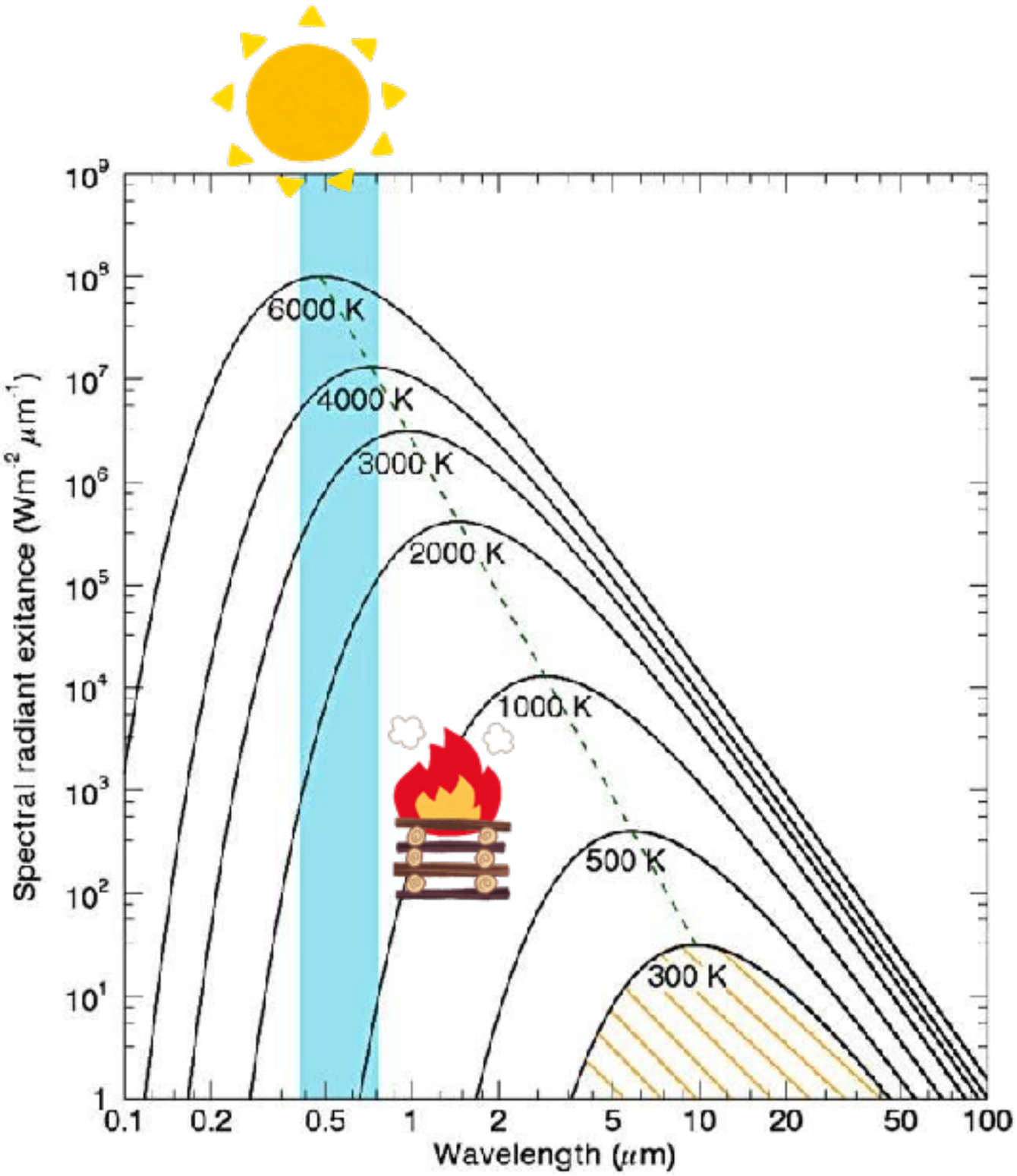
$$E = kT$$

k is Boltzmann constant $8.6171E^{-5}$ eV/K

at room temperature, it is about 25 meV = 0.025 eV

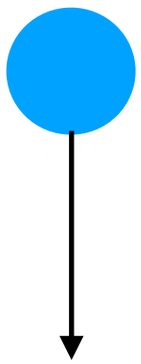
Thermal neutron energy

It is much lower than the ionization energy



https://www.researchgate.net/figure/Blackbody-radiation-curves-at-different-selected-temperatures-as-derived-from-equation_fig17_240856156

Gravitation



Free fall of 1 kg from 1m high

$mgh = \sim 10\ J = 6 \times 10^{19}\ eV = 6 \times 10^{7}\ TeV$

This is a large amount of energy, but it cannot produce high-energy radiation. This is because the energy transferred to each nucleon is very small.

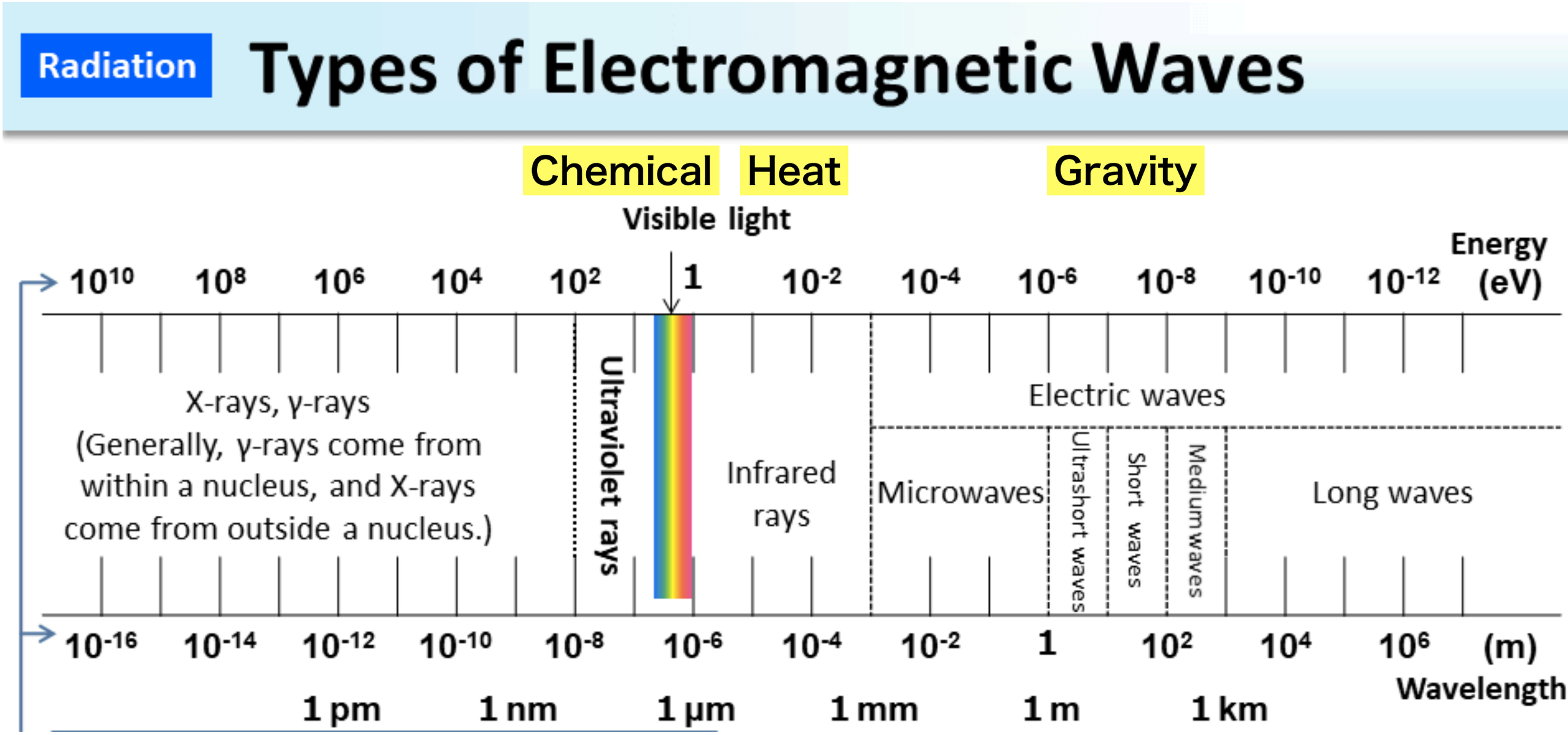
$6 \times 10^{19}\ eV / (1000 \times 6 \times 10^{23}) = 1 \times 10^{-7}\ eV = 100\ neV$

Compared with other energy scales, it is several orders of magnitude smaller

On the surface of neutron stars, “g” is 10^{11} larger

100 neV -> 10 keV !

Ionizing radiation



ionizing radiation

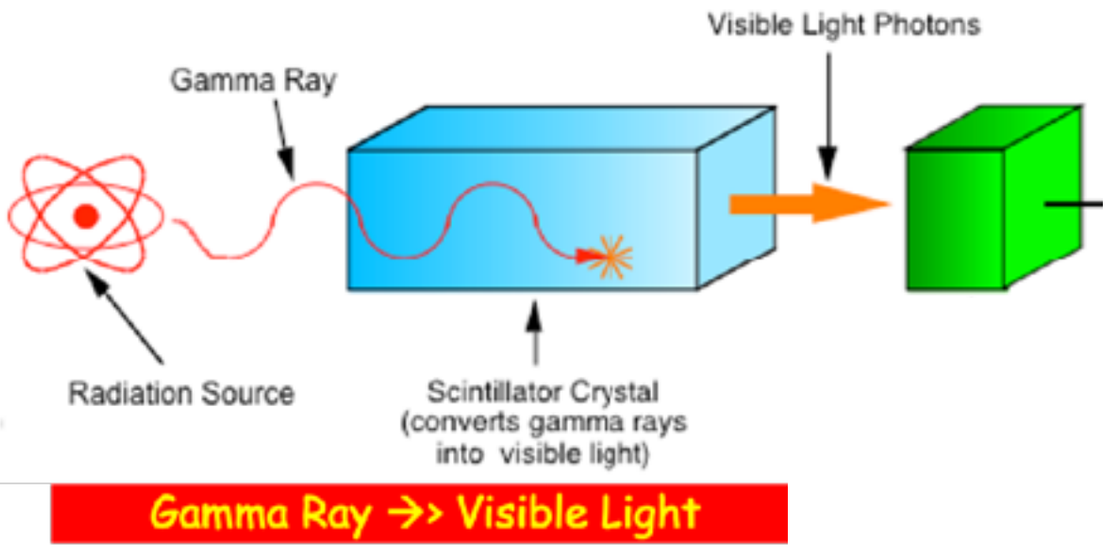
Low-energy quanta cannot ionize an atom.
The total amount of thermal or chemical energy can be very large, but the energy carried by each quantum is too small to cause ionization.

https://www.env.go.jp/en/chemi/rhm/basic-info/1st/index.html

Photo sensor

Photon Counting

How many photons are produced by a scintillator?



case of EasyPET

A LYSO scintillator produces about 16,000 photons for a 511 keV gamma ray. 32 eV gamma energy / one scintillation photon

While scintillation photon energy is 2 eV.

Why are these energies so different?

We see ~one direction out of 6 (L-R,U-D,F-B) direction.

In simplified model, ~2,670 photon will come out.

LYSO is 2mm x 2mm but Sensor is 1mm x 1mm. It is 1/4

Consequently, only about 670 photons will reach the sensor.

The number of photons

How many photon are we seeing?

Visible light is ~2eV

$$1\mathrm{eV} = 1.6 \times 10^{-19}\mathrm{J}$$



$$0.136\ \mathrm{J/s/cm^2}$$

$$= 8.5 \times 10^{17}\mathrm{eV/s/cm^2} = \frac{8.5 \times 10^{17}\mathrm{eV}}{2\ \mathrm{eV/photon}}\mathrm{/s/cm^2} = 4.25 \times 10^{17}\mathrm{photon/s/cm^2}$$

The number of photons

How many photon are we seeing?

Visible light is ~2eV

$$1\mathrm{eV} = 1.6 \times 10^{-19}\mathrm{J}$$

$$1\ \mathrm{J/s} = 6.25 \times 10^{18}\mathrm{eV/s} = \frac{6.25 \times 10^{18}\mathrm{eV}}{2\ \mathrm{eV/photon}}/\mathrm{s} = 3 \times 10^{18}\mathrm{photon/s}$$



0.1s flush ~ 3×10^{17} photon



R



r

Into eye $\frac{\pi r^2}{4\pi R^2} = \left(\frac{r}{2R}\right)^2$ **of probability**

600 photon = 1 W flush 0.1s is seen in 22 km distance

photoelectric effect

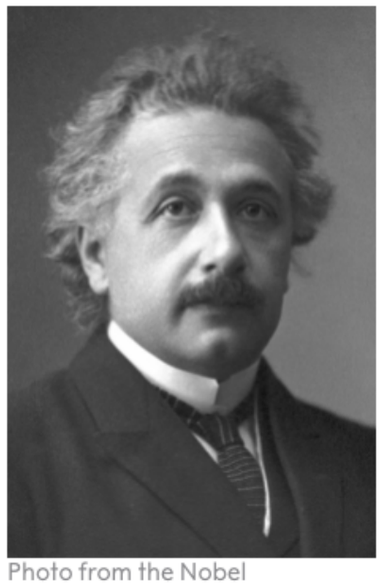
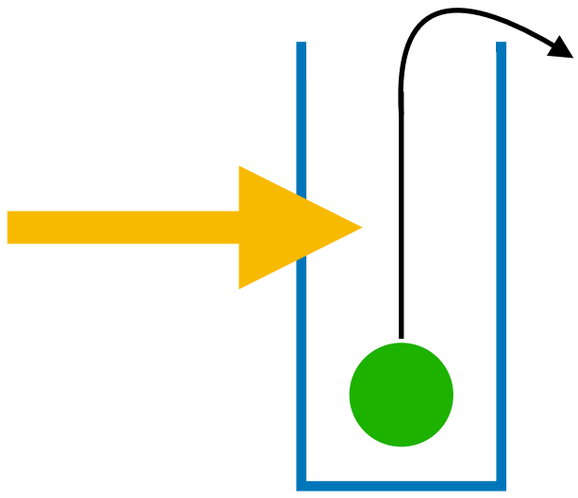
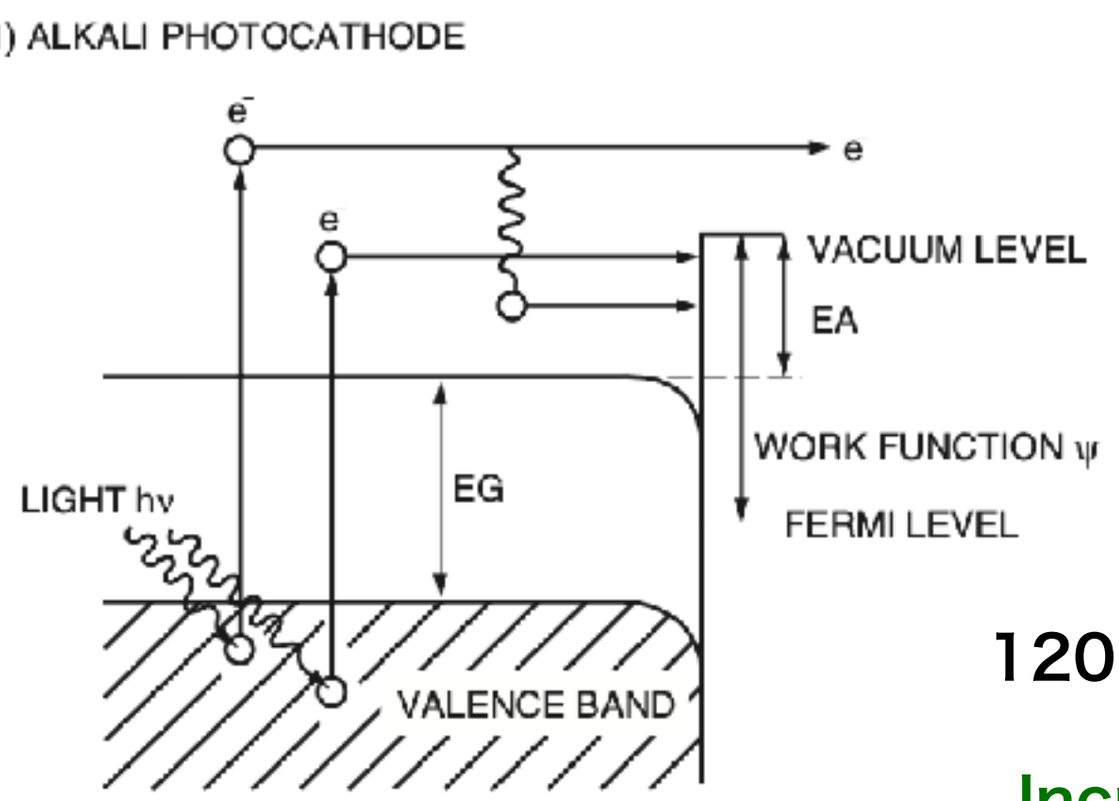


Photo from the Nobel Foundation archive.



Conversion from photon to electron

Charged particles are easier to detect.



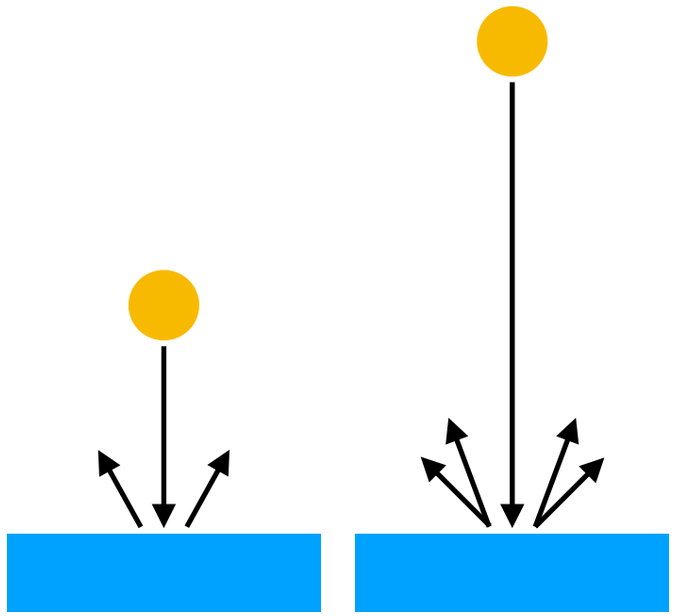
However, efficiency is not 100%.
~ 20 %
600 photons -> 120 electrons

120 electrons are difficult to measure.

Increase the number of electrons !

Electron multiplication

An accelerated electron strikes a metal surface.



Secondary electrons are emitted.

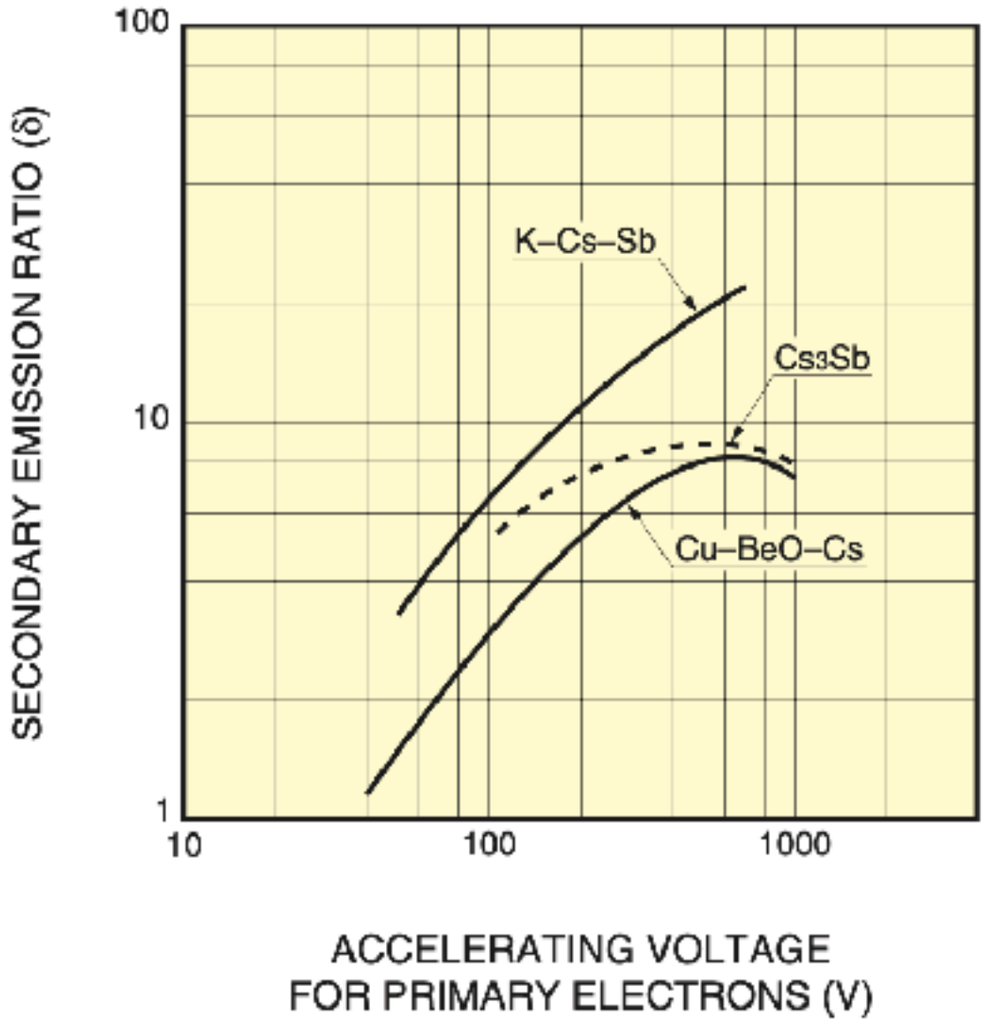
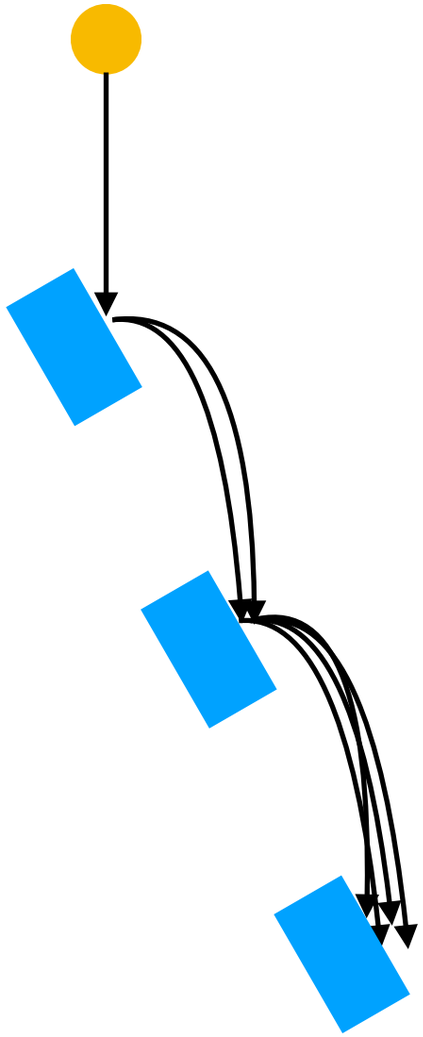
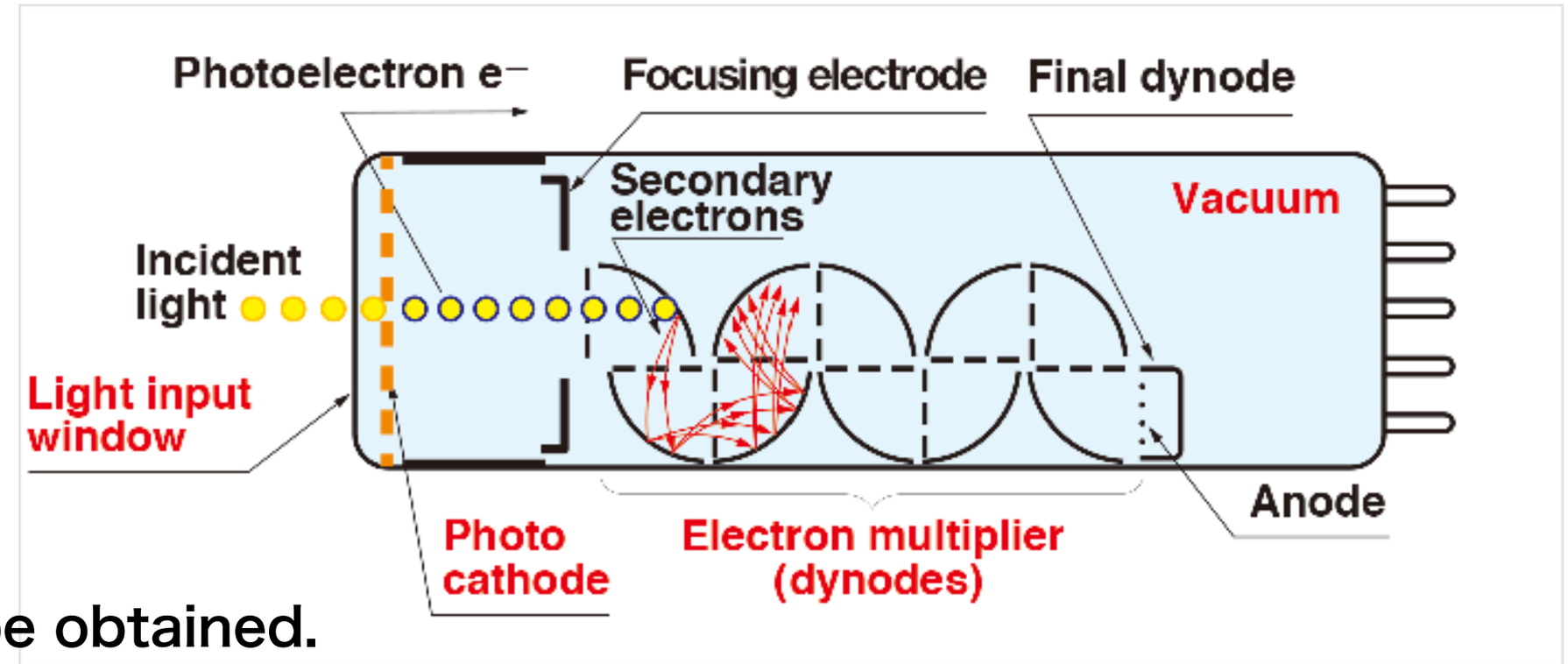


Figure 2-7: Secondary emission ratio

Photomultiplier







Multi-stage

10^5~10^6 gain can be obtained.

120 photoelectrons are multiplied to approximately 10^8 electrons.

PHOTOMULTIPLIER TUBES

Basics and Applications

FOURTH EDITION



HAMAMATSU

PHOTON IS OUR BUSINESS

https://www.hamamatsu.com/content/dam/hamamatsu-photonics/sites/documents/99_SALES_LIBRARY/etd/PMT_handbook_v4E.pdf

Electron Counting

How many electrons do we need to detect?



Ammeter can measure about 1 μA. How many electron?

$$1\mu\mathrm{A} = \frac{10^{-6}}{1.6 \times 10^{-19}}\ \mathrm{e/s}\ = 6 \times 10^{12}\ \mathrm{e/s}$$

We can see→ 0.1s

6 x 10^{11}electron is needed to “See”

10^{8} is not enough to “See”

10^8 electrons pass in 10ns.

Current = Charge/ Time

$1.6 \times 10^{-19} \times 10^8 / (10 \times 10^{-9}) = 1.6$ mA

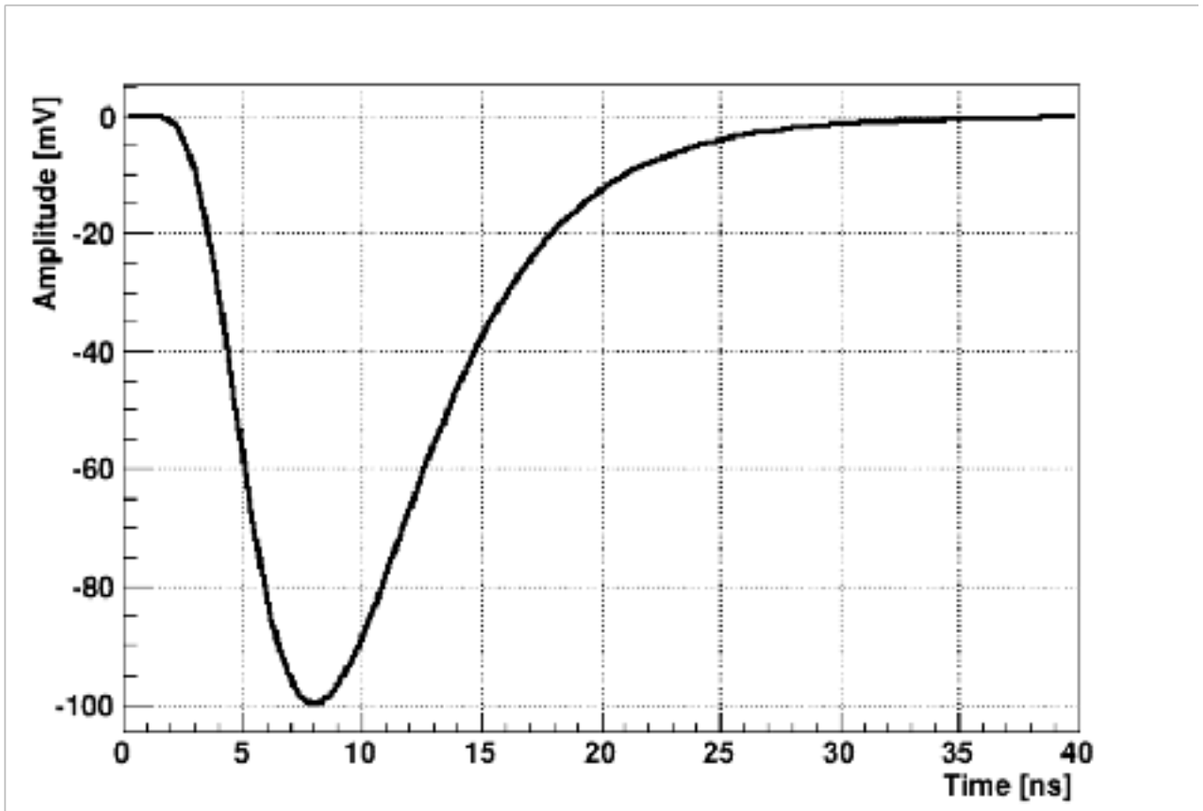
This cannot be measured with an ammeter but can be detected electronically.

Electronics can measure voltage.

Current to Voltage

1.6mA x 50Ω = 80mV

Why 50Ω？

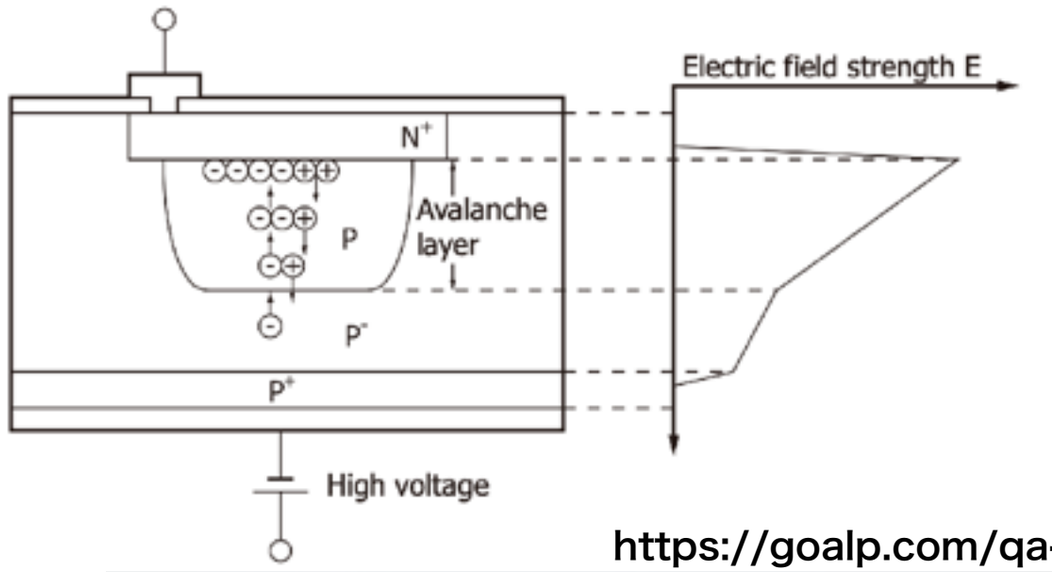


MPPC (SiPM)

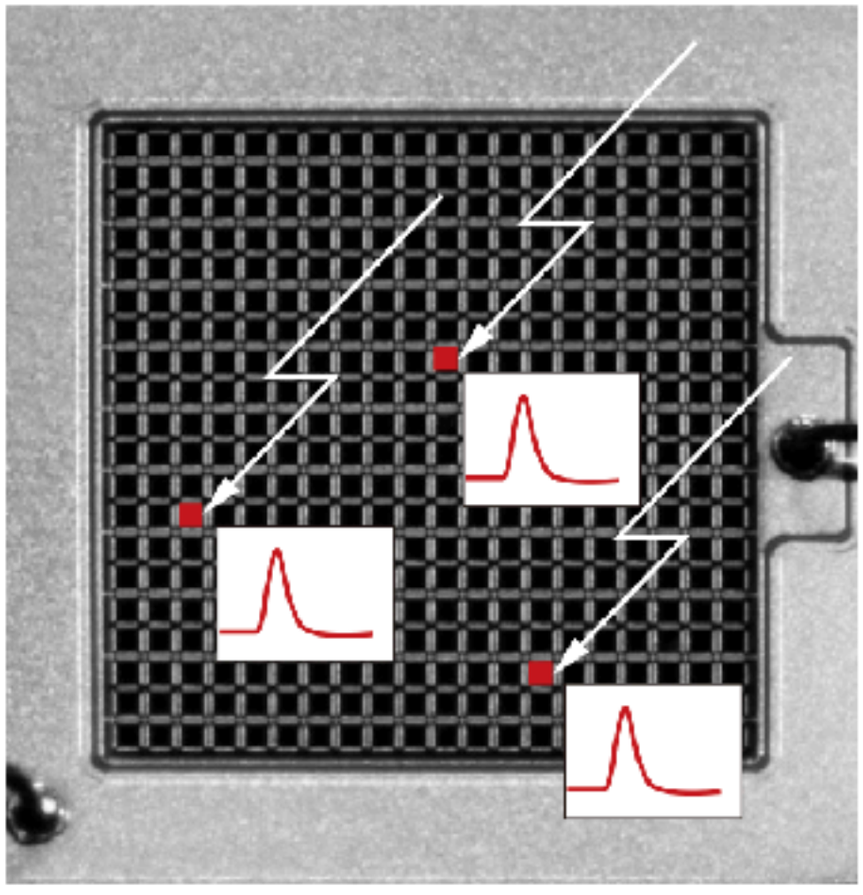
Multi-Pixel Photon Counter (MPPC), also known as silicon photomultiplier (SiPM)

https://www.hamamatsu.com/content/dam/hamamatsu-photonics/sites/documents/99_SALES_LIBRARY/ssd/si-apd_kapd9007e.pdf
https://www.hamamatsu.com/content/dam/hamamatsu-photonics/sites/documents/99_SALES_LIBRARY/ssd/mppc_kapd9008e.pdf

[Figure 1-1] Schematic diagram of avalanche multiplication (near infrared type)



[Figure 1-2] Image of MPPC's photon counting

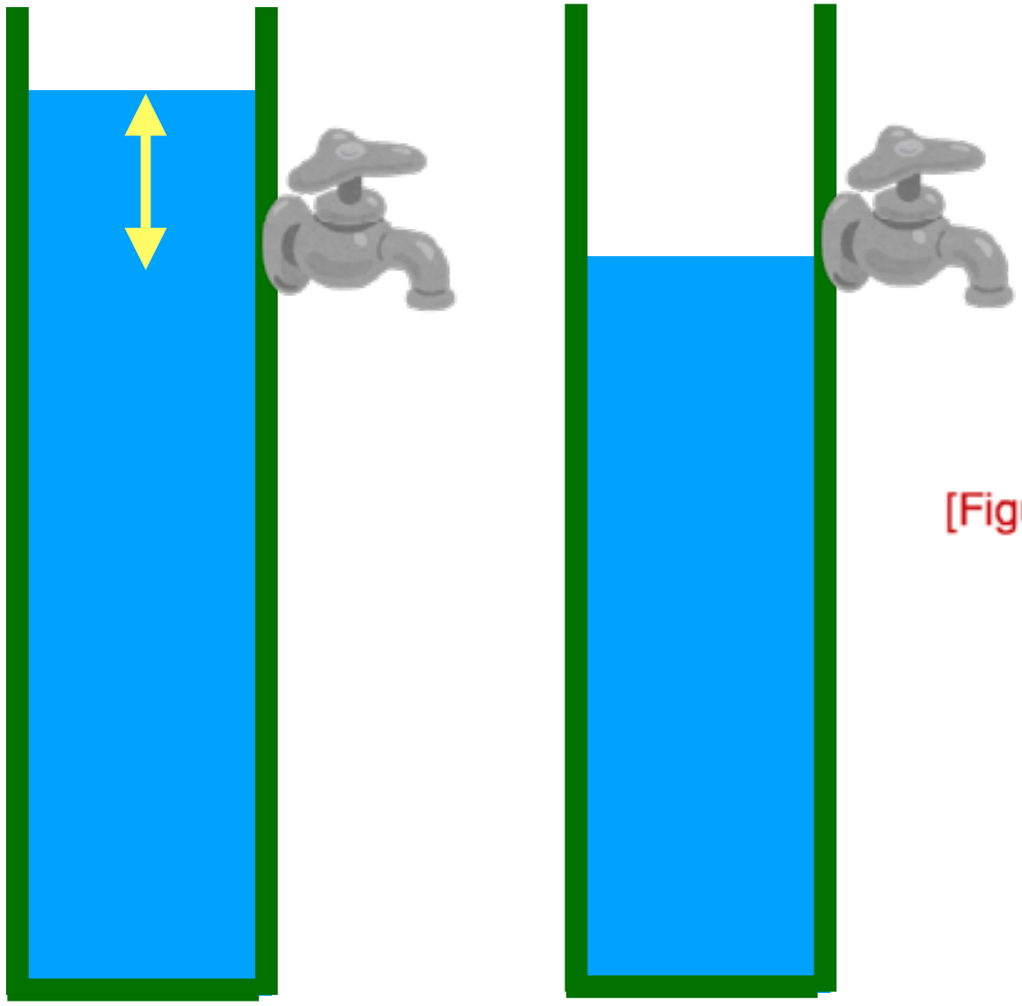


KAPDC0049EA

https://goalp.com/qa-139

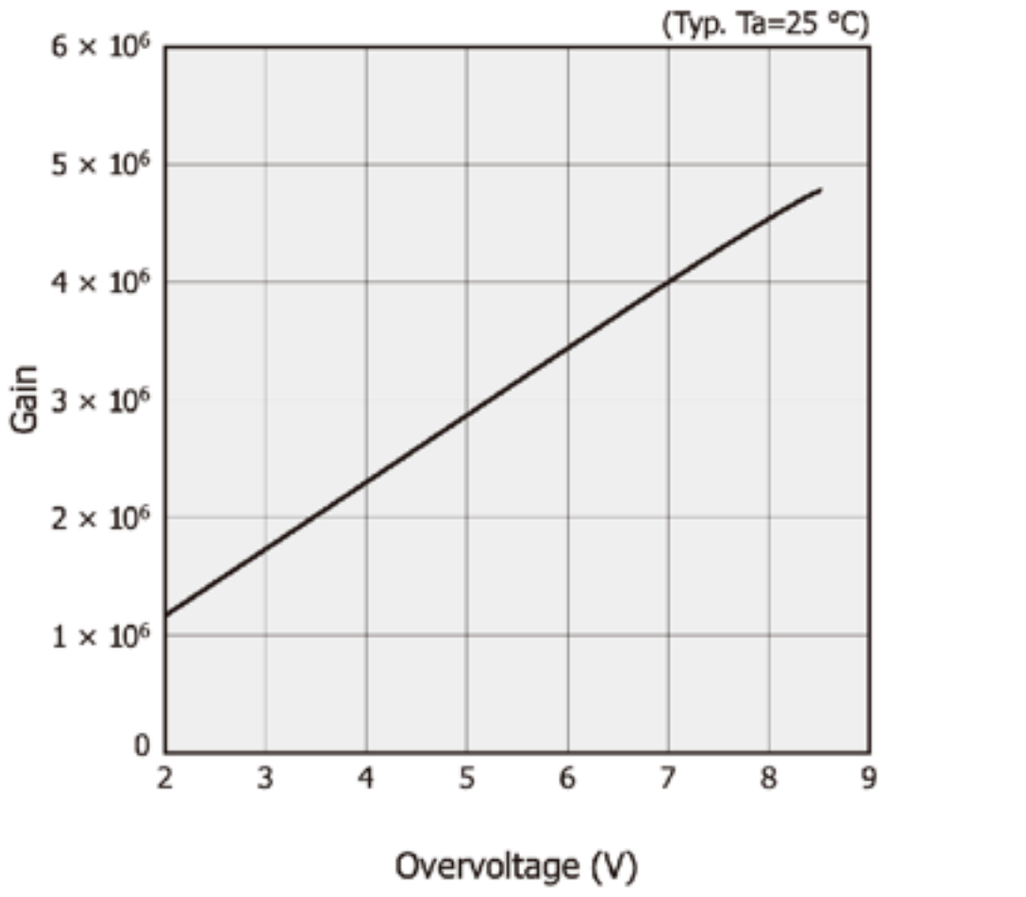


overvoltage



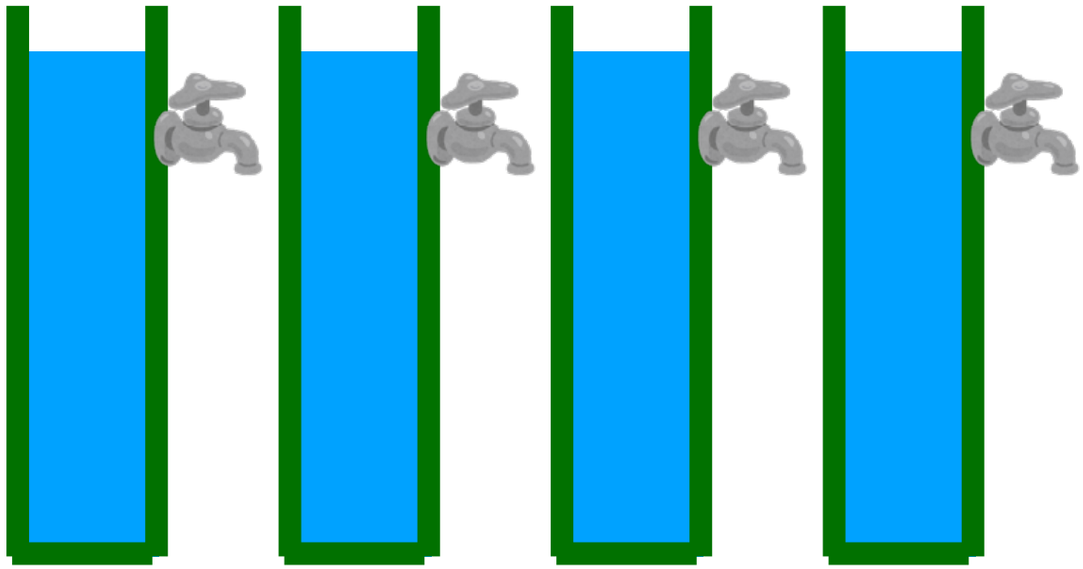
$$Gain = \frac{C\ \Delta V}{q_e},$$

[Figure 3-1] Gain vs. overvoltage (pixel pitch: 50 μm)



Excessively high gain (large avalanches) can damage the device.

Amount of water (Charge) is proportional to the number of hits



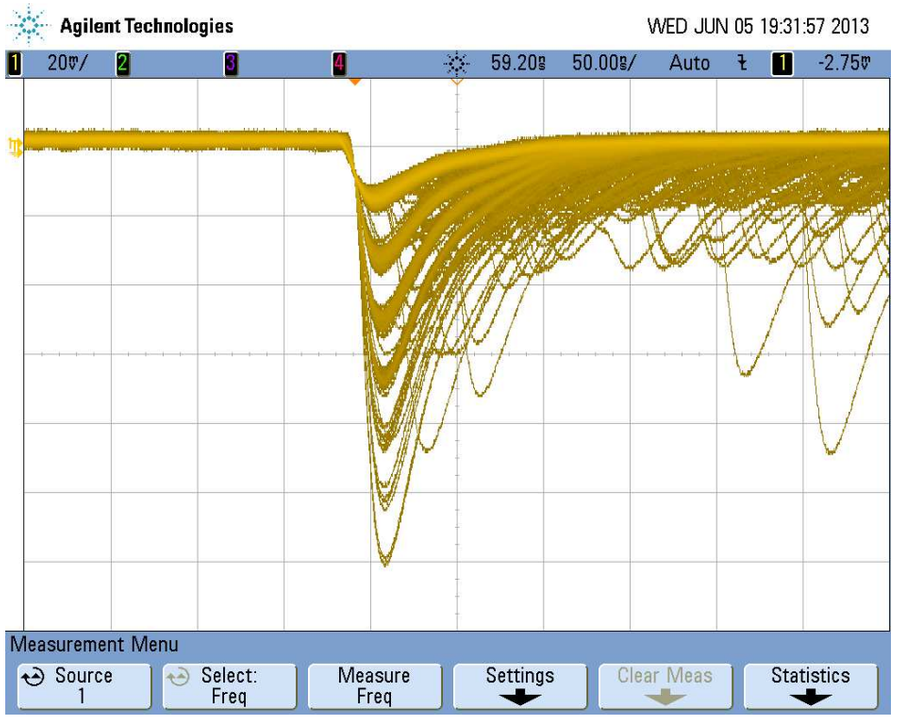
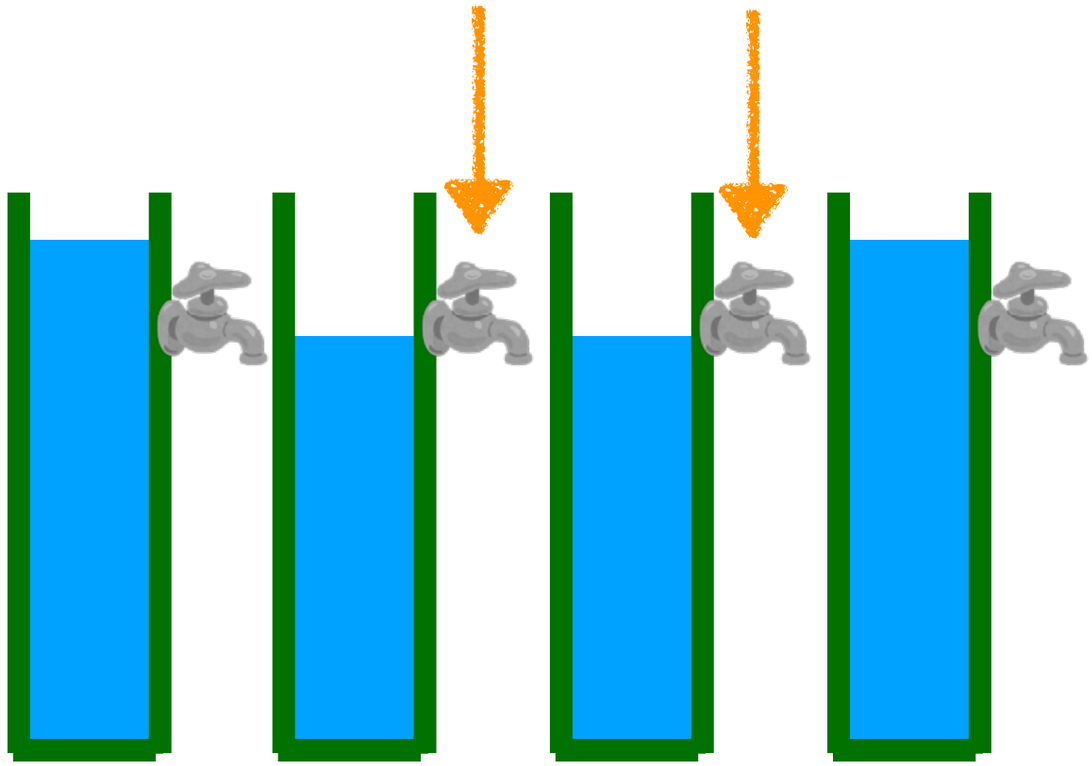
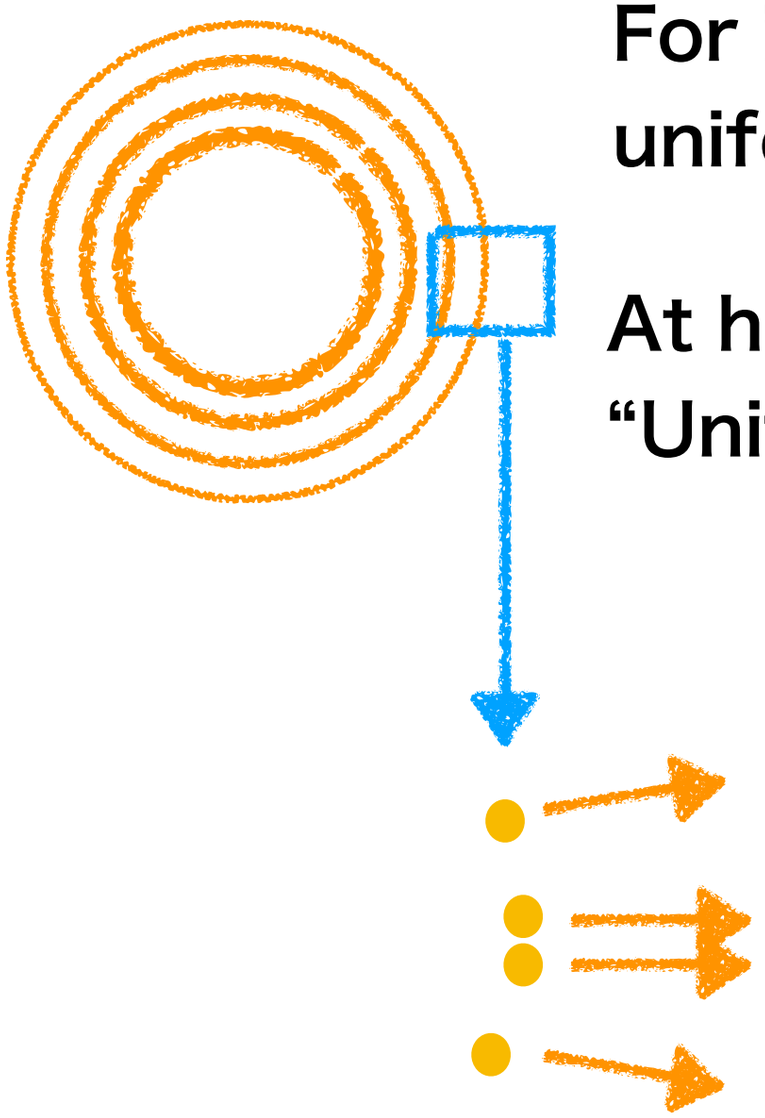


Fig. 1: Response of a SiPM Hamamatsu MPPC S10362-11-100C illuminated by a light pulse.



Statistical fluctuation

Statistical behavior



For isotropic emission, photons are emitted uniformly over solid angle.

At high intensity, the number of photons is large.
“Uniform” means an equal number of photons.

At low intensity, the integer nature of photon counts becomes apparent.

At low intensity, the statistical nature of photons becomes apparent.

“Uniform” means equal probabilities of finding a photon.

From Expected Counts to Observed Counts

The expected counting rate μ is constant,
and not necessary to be an integer.



The observed number of counts must be an integer and fluctuates statistically.



Probability of observing n counts in a time interval t

$P_n(t)$

Poisson distribution

Probability of observing n counts in a time interval t $P_n(t)$

$$P_n(t+\Delta t) = (\mu \Delta t)\, P_{n-1}(t) + (1-\mu \Delta t)\, P_n(t)$$

Skipping the derivation

$$P_n(t) = \frac{(\mu t)^n}{n!} e^{-\mu t}$$

Poisson distribution

Mean and Variance

$$
\begin{aligned}
< n > &= \sum_{n=0}^{\infty} n P_n(t) \\
&= \sum_{n=0}^{\infty} n \frac{(\mu t)^n}{n!} e^{-\mu t} = \mu t \sum_{n=1}^{\infty} \frac{(\mu t)^{n-1}}{(n-1)!} e^{-\mu t} = \mu t
\end{aligned}
$$

$$
\begin{aligned}
\sigma^2 &= \sum_{n=0}^{\infty} (n - < n >)^2 P_n(t) = \sum_{n=0}^{\infty} n^2 P_n(t) - < n >^2 = < n^2 > - < n >^2 \\
&= \mu t(\mu t + 1) - (\mu t)^2 = \mu t
\end{aligned}
$$

$$
< n^2 > = \sum_{n=0}^{\infty} n^2 \frac{(\mu t)^n}{n!} e^{-\mu t} = \mu t \sum_{n=1}^{\infty} n \frac{(\mu t)^{n-1}}{(n-1)!} e^{-\mu t} = \mu t(\mu t + 1)
$$

The mean and variance are equal for a Poisson distribution.

Normal distribution

When the mean of a Poisson distribution is large,

$$\begin{aligned} \log P_n(t) &= n\log(\mu t) - \log(n!) - \mu t \\ &\approx n\log(\mu t) - n\log(n) + n - \mu t \end{aligned}$$

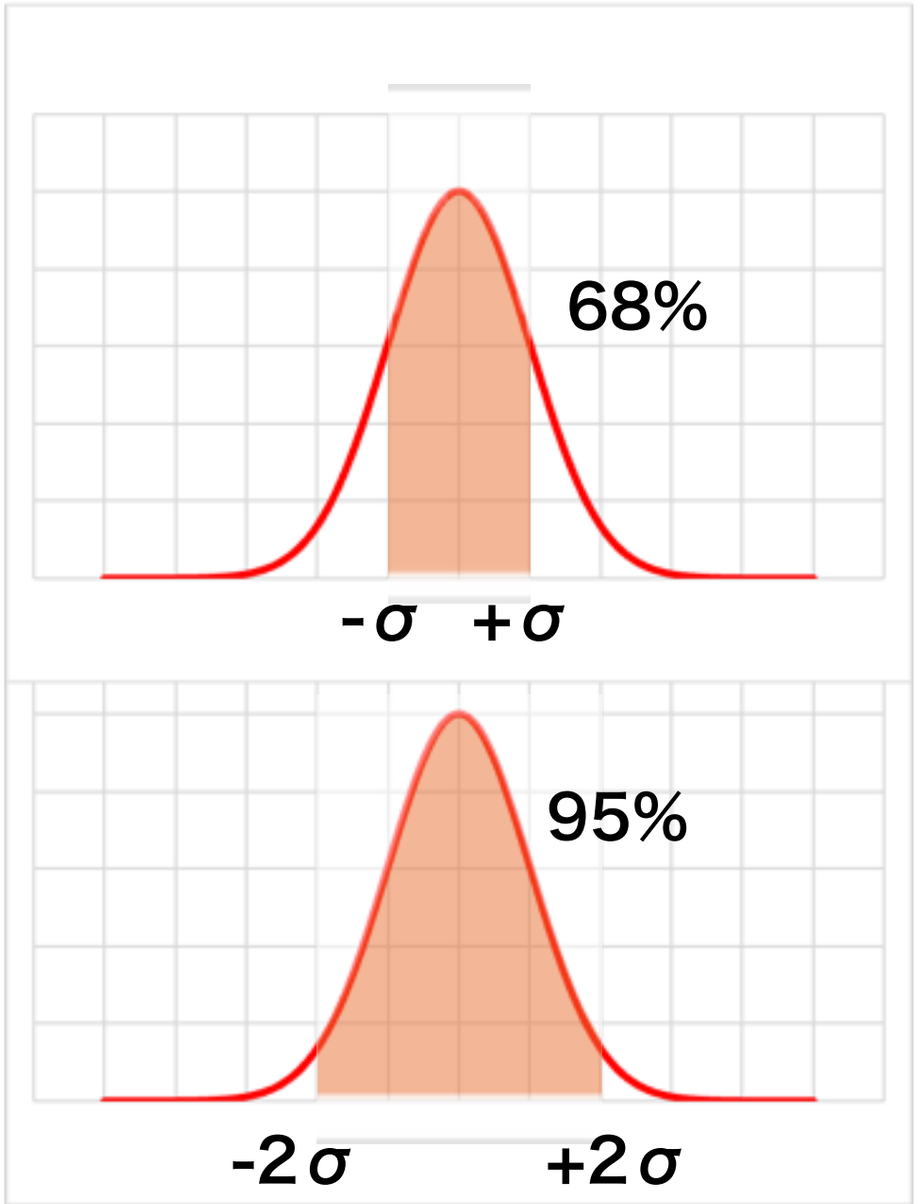
This becomes zero at n=μt.

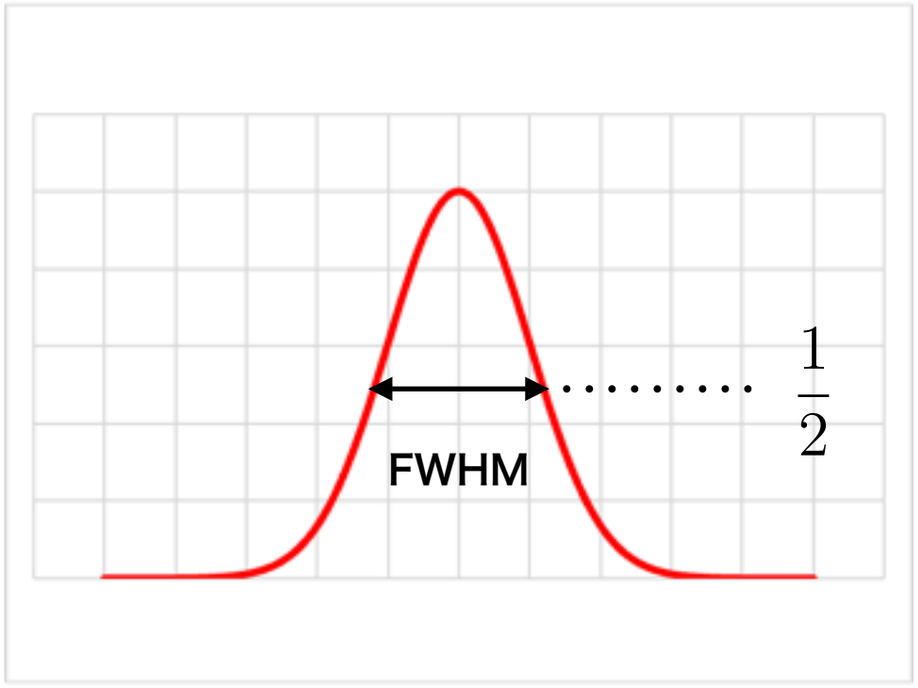
Let's expand this in a Taylor series around n=μt

$$\begin{aligned} \log P_n(t) &\approx -\frac{(n-\mu t)^2}{2\mu t} \\ P_n(t) &\approx e^{-\frac{(n-\mu t)^2}{2\mu t}} \end{aligned}$$

This is known as the normal distribution (also known as the Gaussian distribution) with mean μt and variance μt.

Normal distribution





$$\frac{1}{2} = e^{-\frac{x^2}{2\sigma^2}}$$

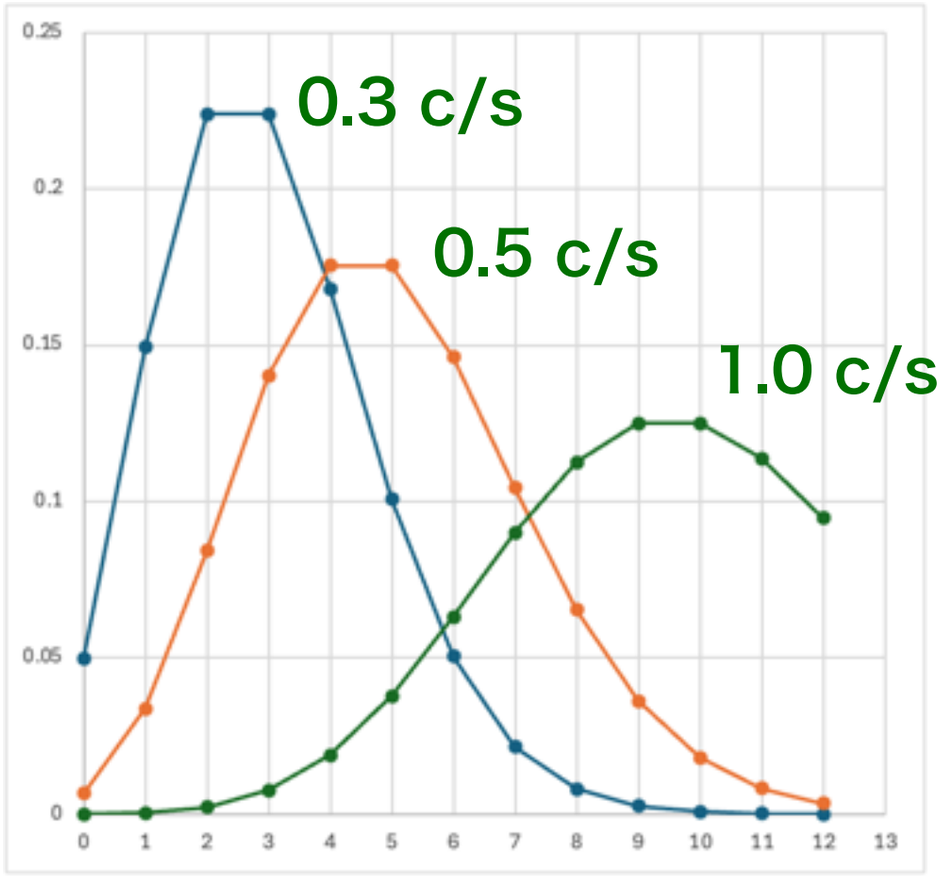
$$x = \sqrt{2\log 2}\sigma$$

$$\mathrm{FWHM} = 2\sqrt{2\log 2}\sigma = 2.35\sigma$$

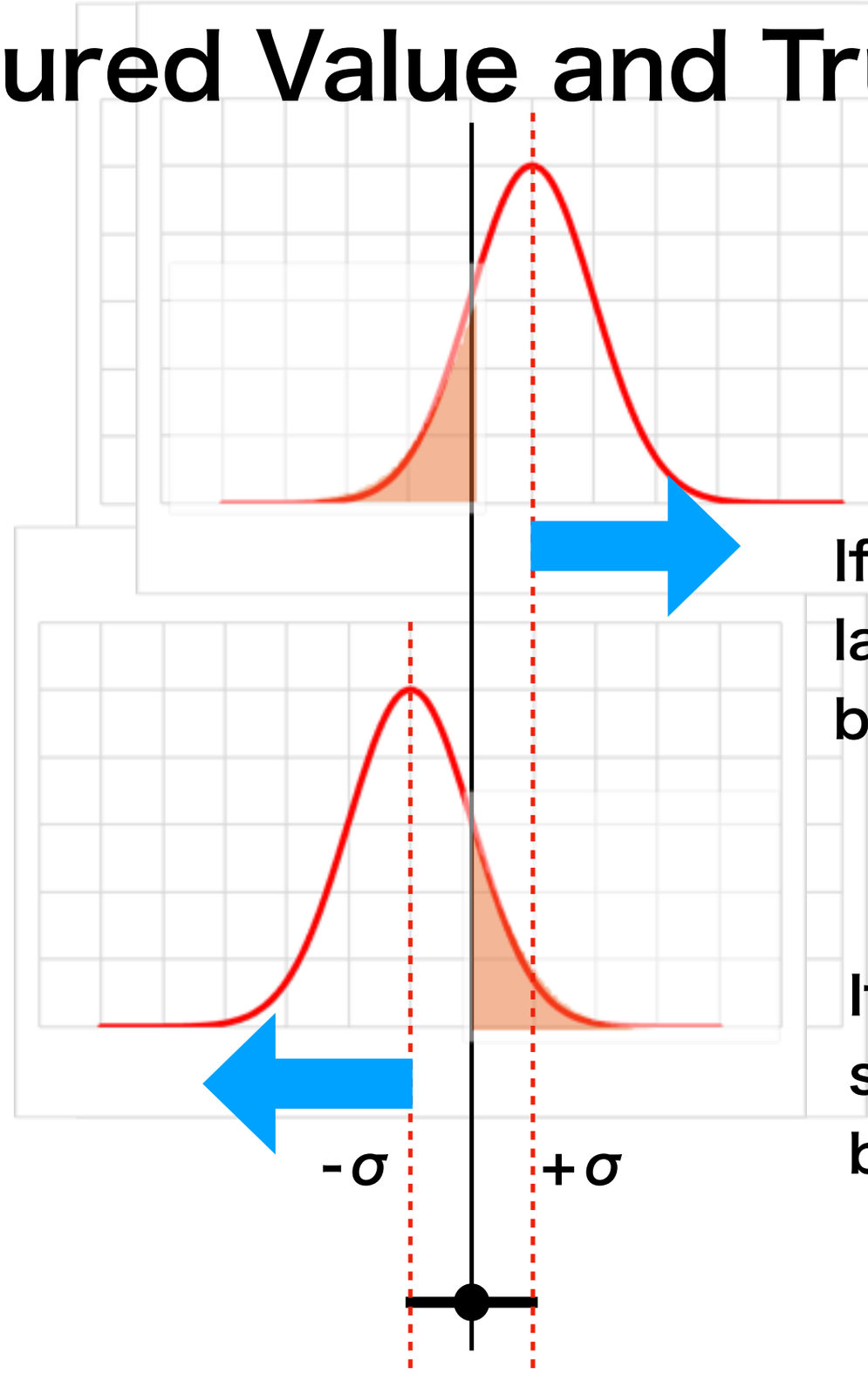
Measured Counts and Counting Rate

If we observe 3 counts in 10 seconds, what is the true counting rate?

0.3 counts/s is a possible counting rate.
0.5 counts/s is also possible.
What about 1.0 counts/s?



Measured Value and True Value



If the true value is much larger, the observed value becomes less probable.

If the true value is much smaller, the observed value becomes less probable.

Error bar

Error Propagation

z = z(x,y) The fluctuations in x and y are independent.

$$\Delta z = \frac{\partial z}{\partial x}\Delta x + \frac{\partial z}{\partial y}\Delta y$$

$$< \Delta z^2 > = < (\frac{\partial z}{\partial x}\Delta x)^2 > + < 2\frac{\partial z}{\partial x}\Delta x\frac{\partial z}{\partial y}\Delta y > + < (\frac{\partial z}{\partial y}\Delta y)^2 >$$

$$\sigma_z^2 = (\frac{\partial z}{\partial x}\sigma_x)^2 + (\frac{\partial z}{\partial y}\sigma_y)^2$$

$$z = x + y$$
$$\sigma_z^2 = \sigma_x^2 + \sigma_y^2$$

$$z = x - y$$
$$\sigma_z^2 = \sigma_x^2 + \sigma_y^2$$

$$z = \frac{x}{y}$$
$$\sigma_z^2 = \frac{1}{y^2}\sigma_x^2 + \frac{x^2}{y^4}\sigma_y^2$$
$$= \frac{z^2}{x^2}\sigma_x^2 + \frac{z^2}{y^2}\sigma_y^2$$
$$\frac{\sigma_z^2}{z^2} = \frac{\sigma_x^2}{x^2} + \frac{\sigma_y^2}{y^2}$$

Hamadohri Environmental Radiation Measurements international school
July 2027

School in English

The Hamadohri International school can be found at
https://indico.rcnp.osaka-u.ac.jp/e/Hamadohri-International-2025
https://indico.rcnp.osaka-u.ac.jp/e/Hamadohri-International-2026

The school will be held again in 2027.
https://indico.rcnp.osaka-u.ac.jp/e/Hamadohri-International-2027

Masaharu Nomachi

Research Center for Nuclear Physics
University of Osaka
nomachi@rcnp.osaka-u.ac.jp